
Introduction to AI

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Notice

- Slides prepared by Moayad Alnammi – malnammi.github.io
- Some figures adapted from CS188: Introduction to Artificial Intelligence, UC Berkeley (Dan Klein, Pieter Abbeel) — ai.berkeley.edu
- Some figures are obtained from the [AIMA Book](#)
- Some content adapted from Dr. ElSayed El-Alfy

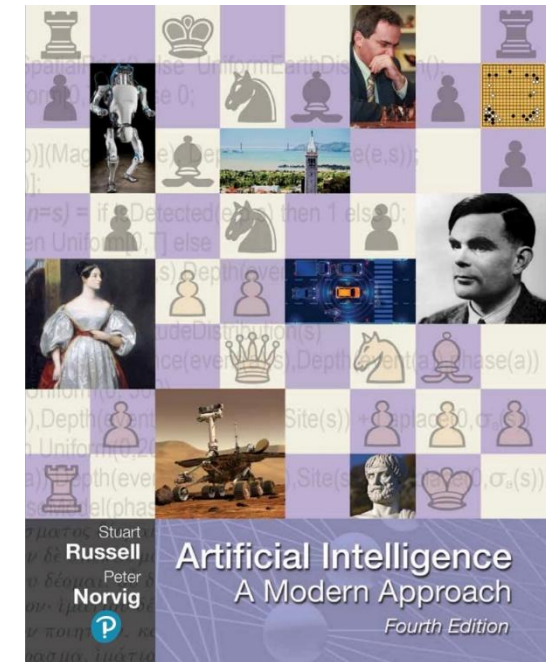
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Outline

- Course Info
- What is AI?
- Foundations of AI
- Quick history of AI
- Current state of AI
- Summary

Course Info

- Instructor: Moayad Alnammi مؤيد النعمي
- Blackboard will be main source of content distribution and assignment submissions.
- Main textbook: Artificial Intelligence: A Modern Approach, 4th ed. [Textbook website](#)
- Syllabus discussion:
 - Grade breakdown
 - Topics list
 - Policies



Why learn AI? and Applications of AI

Why learn AI?

- AI today is everywhere and used to solve many complex problems.
- Automation
- Information Retrieval (e.g. intelligent search engines)
- Fast and effective medical diagnosis
- Operating in risky and complex environments
 - e.g. robots cleaning houses; assisting patients; working in nuclear plants, etc
- Data mining to look for interesting patterns
- Enhancing training through use of simulation
- Stock market analysis, weather forecasting, oil exploration, etc
- Let's see some applications of AI.

Robotics

- Combination of Mechanical Engineering and AI.
- Autonomous vehicles. Waymo Pacifica driverless minivan shown
- Autonomous tracking drones
- Robot soccer RoboCup
- Humanoid robots. ASIMO (Honda) and ATLAS (Boston Dynamics) shown.
- **Note:** in this class we do not cover robotics/hardware aspect of AI. Focus is on the logic/decision-making/software aspect.
- See this concentration if you are interested in robotics:
<https://cx.kfupm.edu.sa/cx/68>



Robot Assistant – Folding clothes [[video link](#)]



Robot Parkour [[video link](#)]

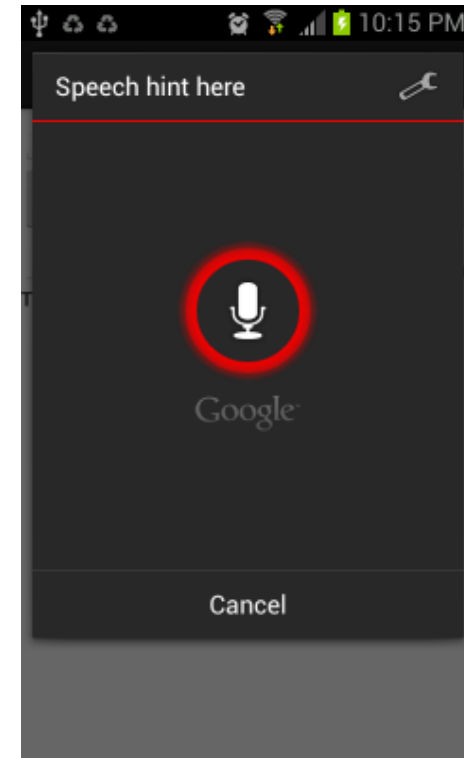
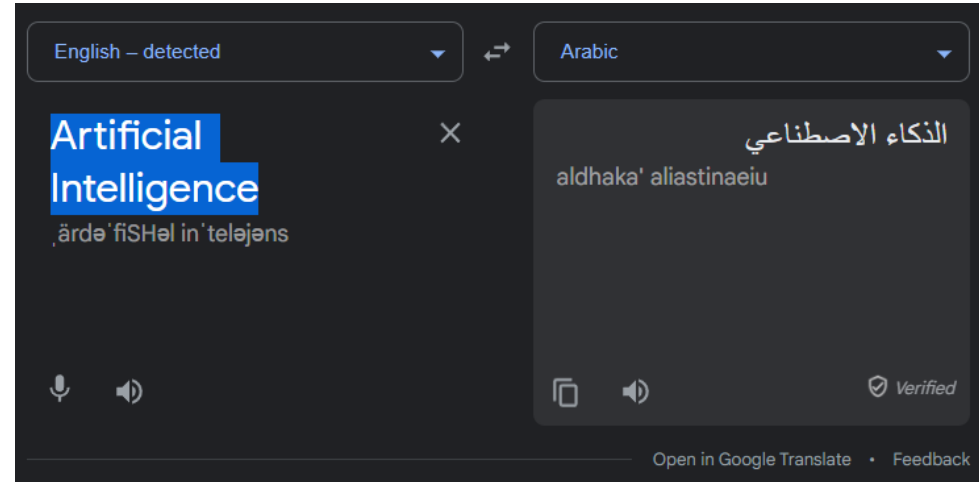


Self-driving car [[video link](#)]



Natural Language

- Speech technologies (e.g. Siri)
 - Automatic speech recognition (ASR)
 - Text-to-speech synthesis (TTS)
- Language processing technologies
 - Machine translation
 - Human-like text (e.g. ChatGPT)



Poor English input: Mary arranged for, in St. Louis, John to rent a house cheap.

Good English output: Mary arranged for John to rent a house in St. Louis.

Poor English input: Today I have went to the store to to buys some many bottle of water.

Good English output: Today I went to the store to buy some bottles of water.

Poor English input: I have tried to hit ball with bat, but my swing is has miss.

Good English output: I tried to hit the ball with the bat, but my swing missed.

Human-like text [ChatGPT](#)

You
Khalid is the son of Ahmed. Mohammed is the son of Ahmed. Waleed is the brother of Ahmed. What is the relationship between Mohammed and Waleed?

ChatGPT
Given the information provided:

- Khalid is the son of Ahmed.
- Mohammed is the son of Ahmed.
- Waleed is the brother of Ahmed.

Since both Mohammed and Khalid are sons of Ahmed, and Waleed is the brother of Ahmed, this implies that Mohammed and Waleed share the same father, making them brothers as well. Therefore, Mohammed and Waleed are brothers.



Can make logical mistakes. Although it replies confidently.

You
Write an Arabic poem about bravery

ChatGPT
ببساطة الروح ننير الطريق
في عتمة الليل وبظلمات الخطى

الشجاعة تملأ القلب بالنور
تحمل في جعبتها أحلام الشموخ

على جبين الشجاع يتسلق الأمل
كالشمس الباسمة في سماء الصباح

Not bad Arabic output.

Expert/Decision-making/Recommender Systems/knowledge-based

- Search Engines
- Recommender systems
- Spam Filters
- Automated helpdesks
- Medical diagnosis systems
- Automated trading
- Fraud Detection
- Fault Detection



YAHOO!



Aol.

Yandex

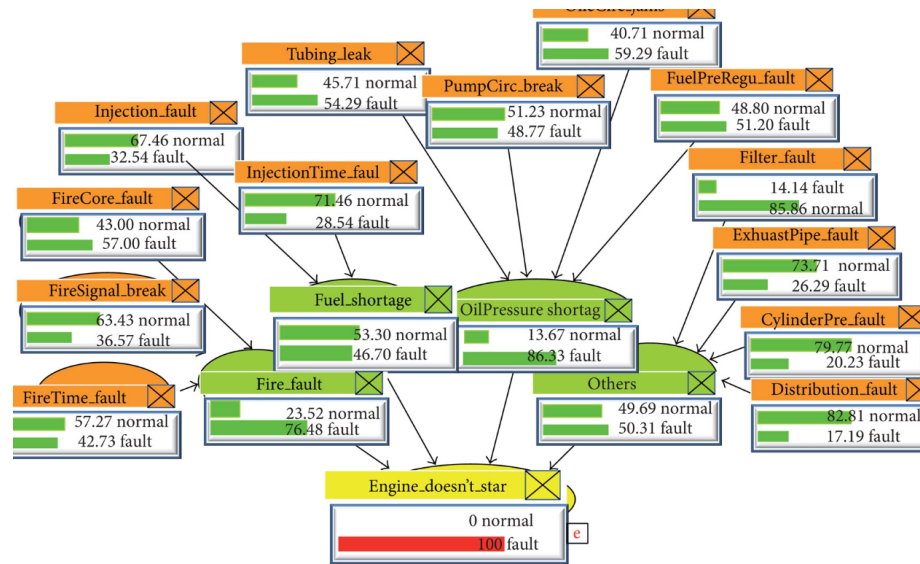
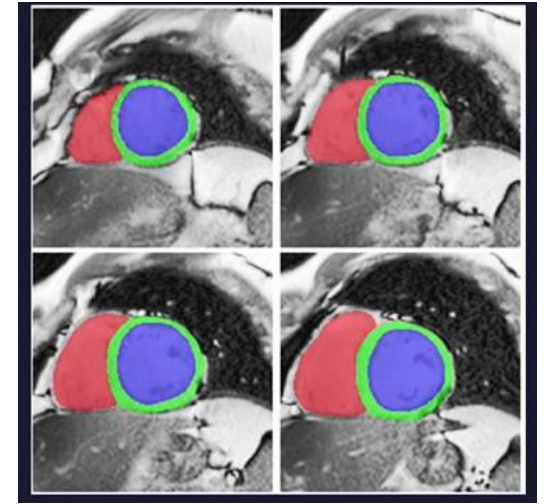


Image from [link](#)

amazon.com

Recommended for You

Amazon.com has new recommendations for you based on [items](#) you purchased or told us you own.



[Google Apps Deciphered: Compute in the Cloud to Streamline Your Desktop](#)



[Google Apps Administrator Guide: A Private-Label Web Workspace](#)



[Googlepedia: The Ultimate Google Resource \(3rd Edition\)](#)

Navigation & Searching

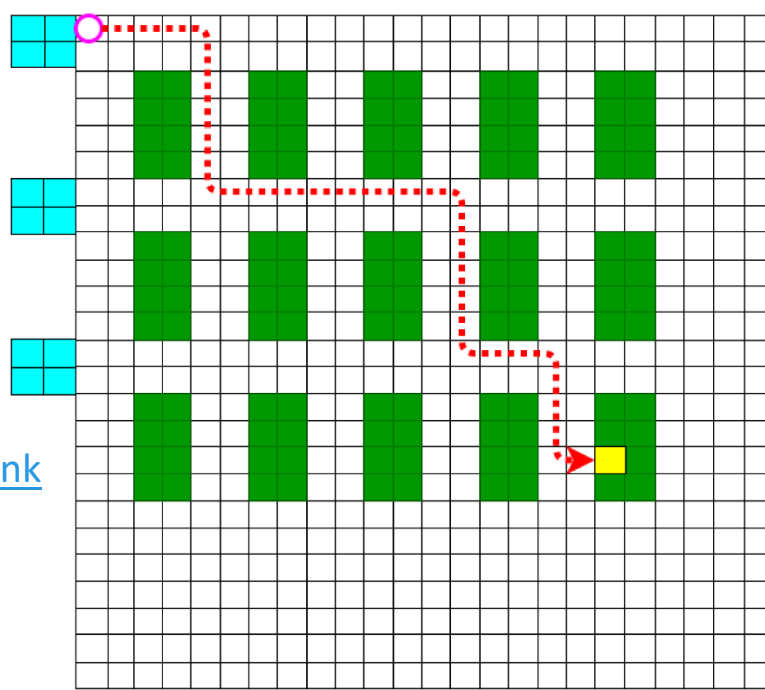


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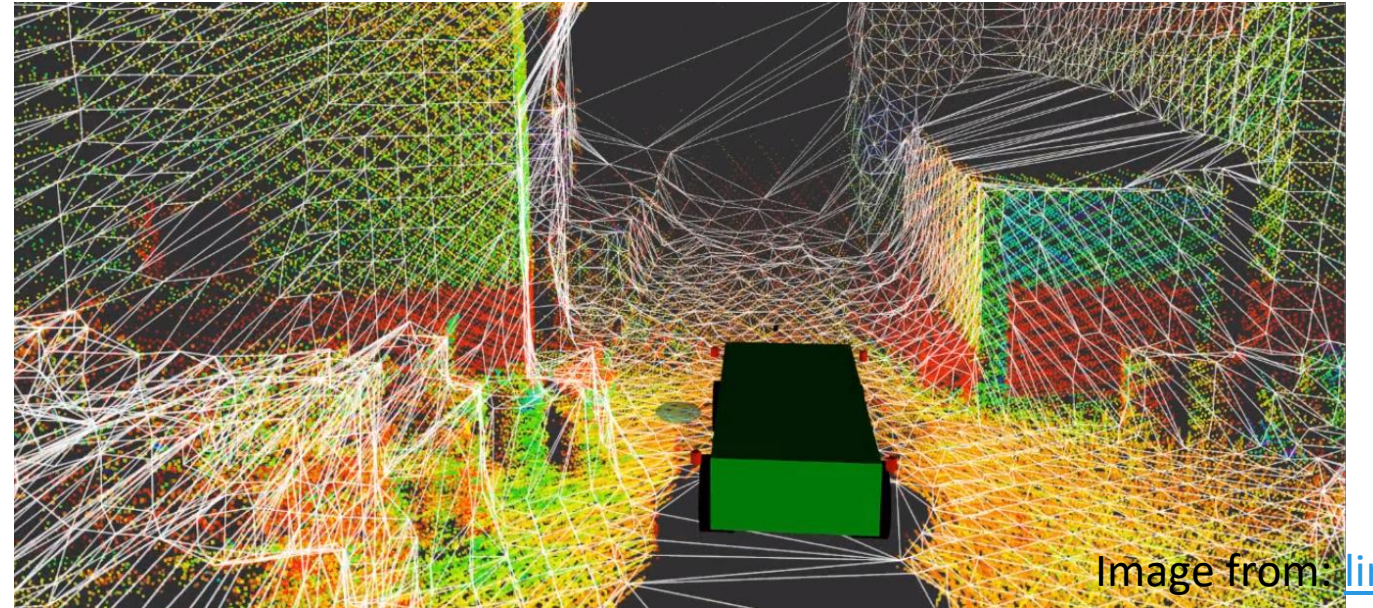
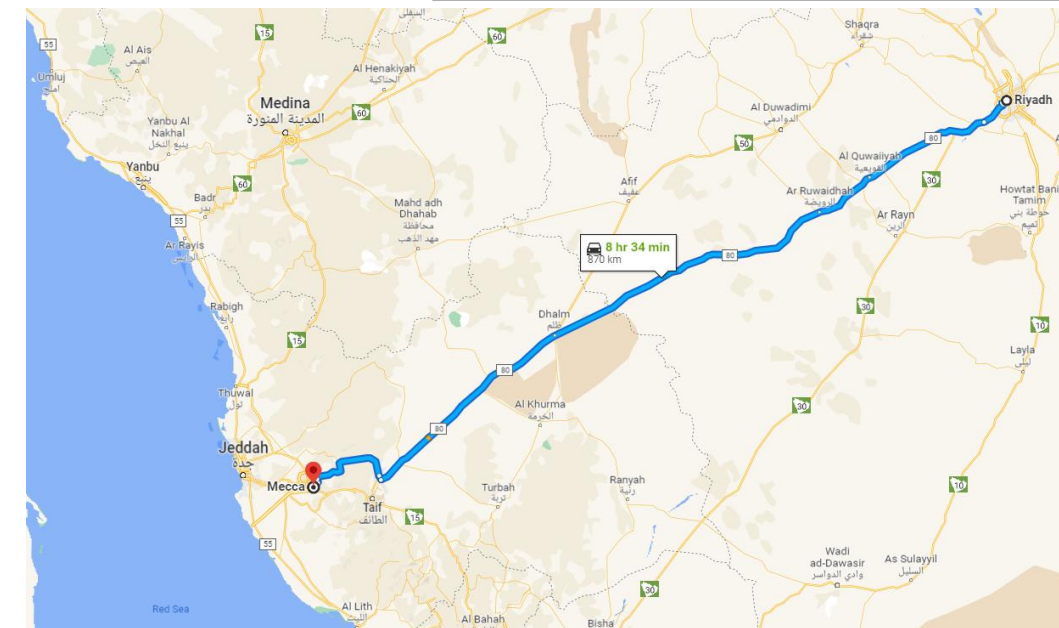
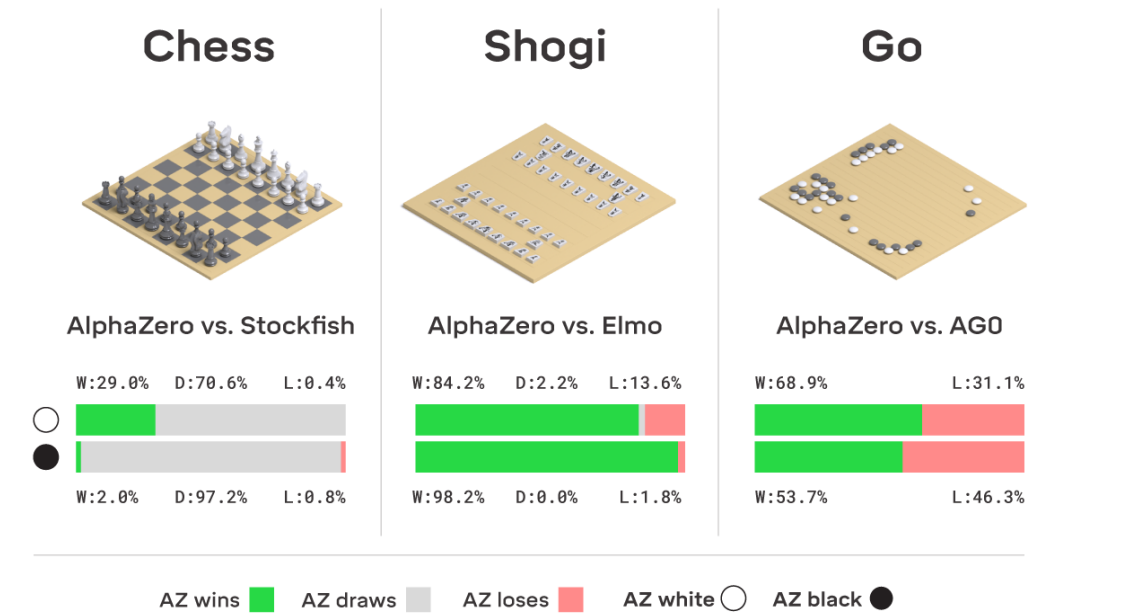


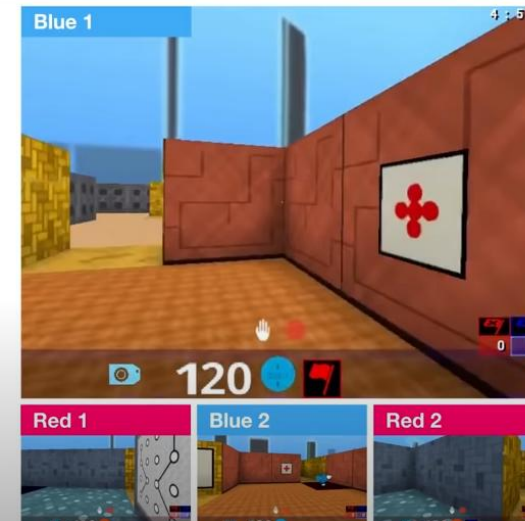
Image from: [link](#)

Game Playing

- DeepBlue defeats chess champion [1997]
- AlphaZero defeats humans and other AI in chess, shogi and go [2018].
- Open AI Five beats Dota champions. [2018]
- [Deepmind's Quake 3 AI](#) [2018].
- [AlphaStar](#): Grandmaster level in StarCraft II [2019].



Agent observation raw pixels



Scheduling/Optimizing

- Finding optimal configuration of a system.
- Job scheduling
- Circuit design
- Cargo packing
- Flight scheduling
- Registrar scheduling

Course Offering

Select Term

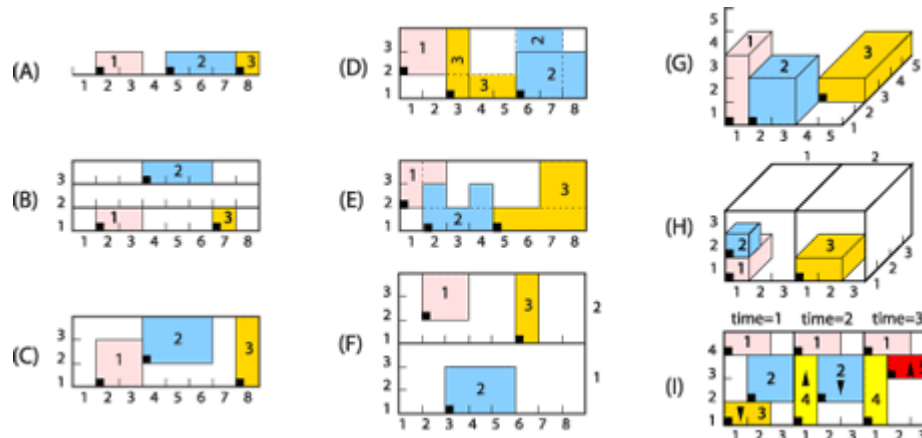
Term 222

Select Department

Accounting & Finance

Course-Sec	Activity	CRN	Course Name	Instructor	Day	Time	Location	Status	Waitlist
ACCT110-01	LEC	21684	Introduction to Financial Accounting	ABDULLAH ALHARBI	MW	1230-1345	24-174	Closed	Open
ACCT110-02	LEC	21083	Introduction to Financial Accounting	ABDULLAH ALHARBI	MW	1400-1515	24-174	Open	Closed
ACCT210-01	LEC	21087	Introduction to Managerial Accounting	MIRGHANI AHMED	UT	1100-1215	24-174	Closed	Open
ACCT210-04	LEC	21090	Introduction to Managerial Accounting	WALID CHEFFI	MW	1230-1345	24-244	Open	Closed
ACCT210-05	LEC	21982	Introduction to Managerial Accounting	MIRGHANI AHMED	UT	1400-1515	24-250	Open	Closed

Course-instructor-room scheduling



Packing problems. Image from [link](#)

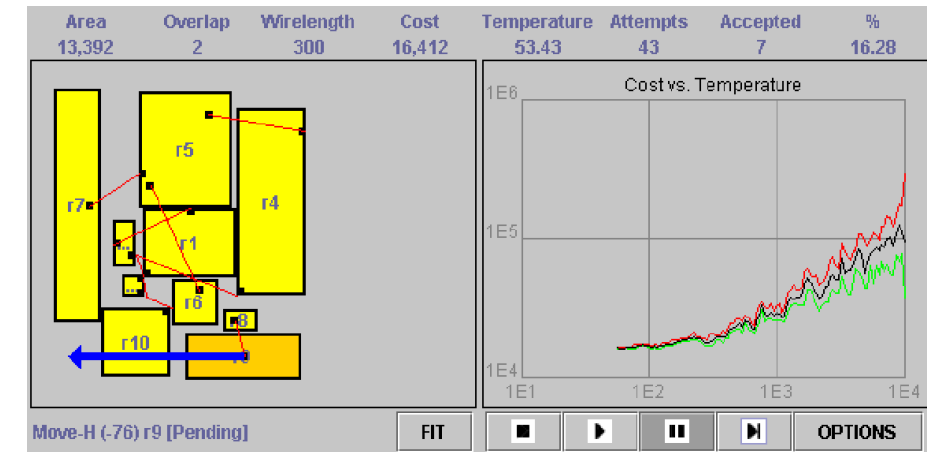


Figure 5 – Simulated Annealing Placement Applet

Circuit design. Image from [link](#)

CS GO Autonomous Bot just from pixels input [[video link](#)]



Paper [link](#)

What is AI?

Discussion

- What is intelligence?
- Can it be learned?
- Can it be taught?
- Can it be measured and quantified?
- Can machines/computers be intelligent?

What is Intelligence?

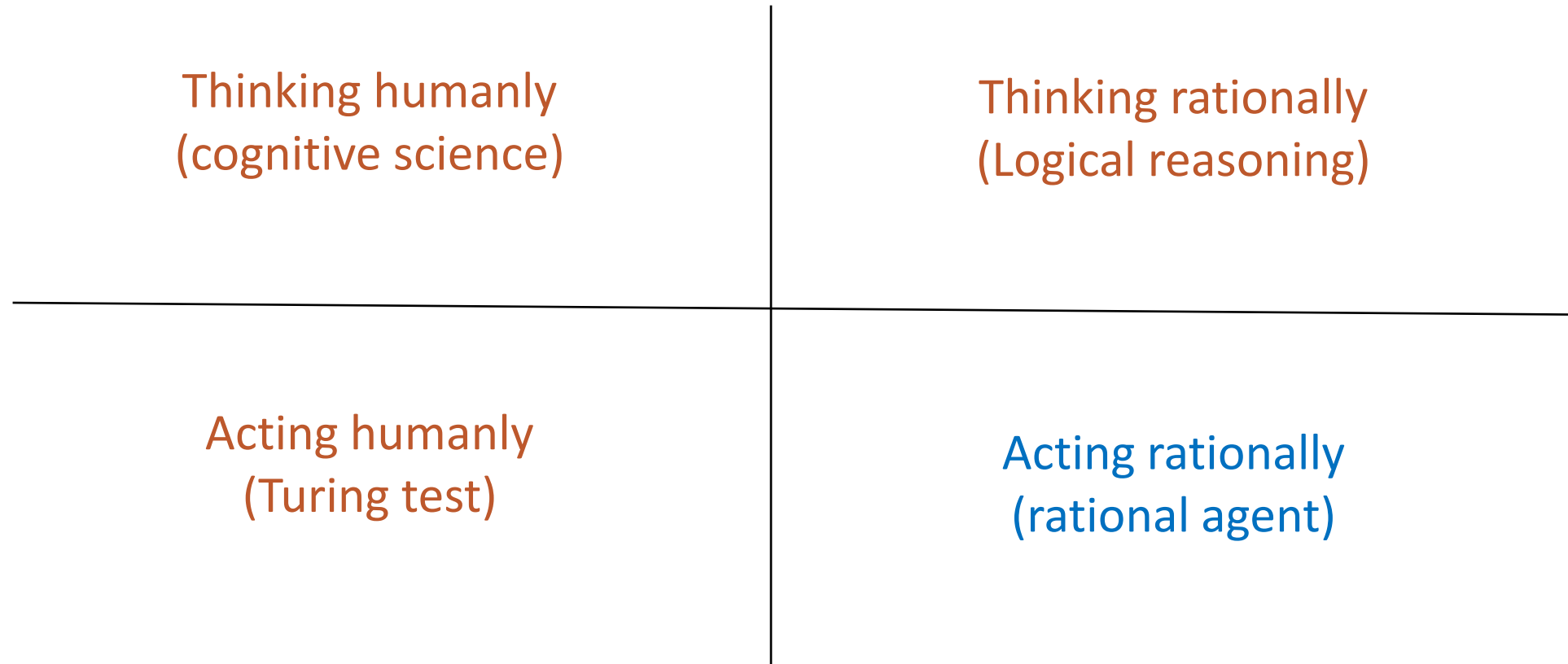
- “(1) the ability to **learn** or **understand** or to **deal** with new or trying **situations** :**reason**; also: skilled use of reasoning” - Merriam Webster
- “(2):the ability to **apply** knowledge to manipulate one's environment or to **think** abstractly as measured by **objective criteria** (as tests)” - Merriam Webster
- “ability to **acquire** and **apply** knowledge and skills” - Oxford

What is AI? – Definition

- Still no crisp complete agreed upon definition. Why?
 - Involves creativity (definition contradicts creativity)
 - Broad multidisciplinary area (many perspectives, many applications, many technologies, etc.)
- According to John McCarthy (the father of AI)
 - Intelligence = **computational** part of the ability to achieve goals in the world
 - “... the **science** and **engineering** of making intelligent machines, especially **intelligent computer programs**”
- Steven Tanimoto : “Computational techniques for performing tasks that apparently require intelligence when performed by humans” Elements of AI, 1995
- George Luger : “... branch of computer science that is concerned with **automation of intelligent behavior**” AI, 2008
- Practical definition: AI is a set of techniques that allow us to build machines that perform tasks.

What is AI? [2]

- According to the textbook AIMA. AI can be defined based on four approaches:



Acting humanly – Turing test approach [1]

- Test proposed by Alan Turing (1950): a human interrogator asks written questions to which a computer responds. The computer passes if the interrogator cannot tell if the responses are coming from a human or a computer.
- The computer would need the following capabilities:
 - **Natural language processing** to communicate.
 - **Knowledge representation** to store what it knows or hears (e.g. memory).
 - **Automated reasoning** to answer the given questions and make conclusions.
 - **Machine learning** to detect patterns and generalize to new situations (predict correctly).
- Extra capabilities (total Turing test):
 - **Computer vision and speech recognition** to see and hear (perception).
 - **Robotics** to move and interact with the world.
- These six disciplines are subfields of AI.
- Is passing the Turing test a good scientific goal?
 - Not a good way to solve practical problems. Too general for specific tasks we care about.
 - Can create intelligent agents without trying to imitate humans.

Thinking humanly – Cognitive modelling approach

- Develop computer programs that model how humans think.
- How to understand cognition as a computational process?
 - Introspection: try to think about how we think
 - Psychological experiments: predict and test behavior of human subjects
 - Image the brain, examine neurological data
- A computer program that mimics the input-output behavior of humans, then there may be evidence that it thinks like a human.
- Cognitive science: blends computer modelling and experimental psychology to develop theories on how the human mind operates.

Thinking rationally – Logical Reasoning “Laws of Thought” approach

- Emphasis on making correct inferences (derive conclusions out of premises).
- Logic: patterns of argument that always yield correct conclusions when supplied with correct premises, e.g.

“Socrates is a man; all men are mortal; hence Socrates is mortal”

- Logician approach to AI: describe problem in **formal logical** notation and apply general deduction procedures to solve it.
- Issues with this approach:
 - Requires **certain (correct)** information. Most problems have a level of **uncertainty**.
 - Not always possible to describe complex real-world problems and knowledge in logical notation.
 - Leads to an **understanding** of how the world works (**rational thought**), but not to intelligent behavior (**rational action**).

Acting rationally – the **rational agent** approach

- An **agent** is an entity that *perceives* and *acts*.
- A **rational agent** is one that acts in order to **maximize its expected utility**. Utility is another term for reward or profit.
- Thus, the focus is on the decisions/actions, not the cognitive process behind them.
- Advantages of this approach:
 - More general than the “Laws of thought” approach. Some rational actions require careful logical reasoning, other rational actions are reflex-based like removing your hand from a hot stove.
 - All skills required for the Turing test are also helpful for acting rationally: NLP, knowledge representation, automated reasoning, machine learning, etc.
 - Rationality can be **mathematically** defined via **objective (utility)** functions. Can even incorporate **uncertainty** in the objective function via probability.
- For these reasons, this approach has prevailed in the field of AI. It is sometimes referred to as the **standard model of AI**.

The rational agent – The standard model of AI

- Thus, the standard model of AI is as follows:

The development of rational agents that select actions in order to maximize an expected utility function. This utility function is specified by the human developer.

- Perfect rationality may not be feasible due to computational constraints; e.g. it takes too long to find the best action. In such a case, we can opt for bounded/limited rationality.
- Recently, this standard model has been adapted to account for uncertain/unknown preferences. E.g. we want the automated car to reach a destination (goal) but it should do so safely (unspecified preference).
- In this class we will look at various problems and how to build rational agents for those problems.

Foundations of AI

AI borrows from a lot of fields

Philosophy	Logic, methods of reasoning, mind as physical system, foundations of learning, language, rationality
Mathematics	Formal representation and proof, algorithms, computation, (un)decidability, (in)tractability, probability
Economics	Utility, decision theory
Neuroscience	How the brain operates
Psychology	How humans perceive, think, and act
Computer Engineering	Building fast computers
Control Theory	Maximizing an objective function under uncertainty
Linguistics	Knowledge representation, grammar, syntax

Short History of AI

Short history of AI

1940-1950: Early days	1943: McCulloch & Pitts: Neural Nets, Boolean circuit model of brain 1950: Turing's "Computing Machinery and Intelligence"
1950-70: Excitement I	1950s: Early AI programs: chess, checkers (RL), theorem proving 1956: Dartmouth summer meeting: "Artificial Intelligence" adopted 1965: Robinson's complete algorithm for logical reasoning General-problem solvers could not computationally scale to "difficult" problems.
1970-90: Knowledge-based approaches (expert systems; rule-based; if-then production rules)	Shift from "general" to domain-specific problem solvers. Leverage domain knowledge to reduce computational complexity. 1969-79: Early development of knowledge-based systems 1980-88: Expert systems industry booms. Million dollars in 1980 to billions in 1988. 1988-93: Expert systems industry busts: "AI Winter" [models did not account for uncertainty and did not learn from experience]
1986 - 2012: Statistical approaches	Resurgence of probability, focus on uncertainty (probability) General increase in technical depth Agents and learning systems "AI Spring"
2012- ____: Excitement II	Availability of big data and shift towards Machine Learning big compute, advances in GPU, deep learning. AI used in many industries

History of Deep Learning [Neural Networks]

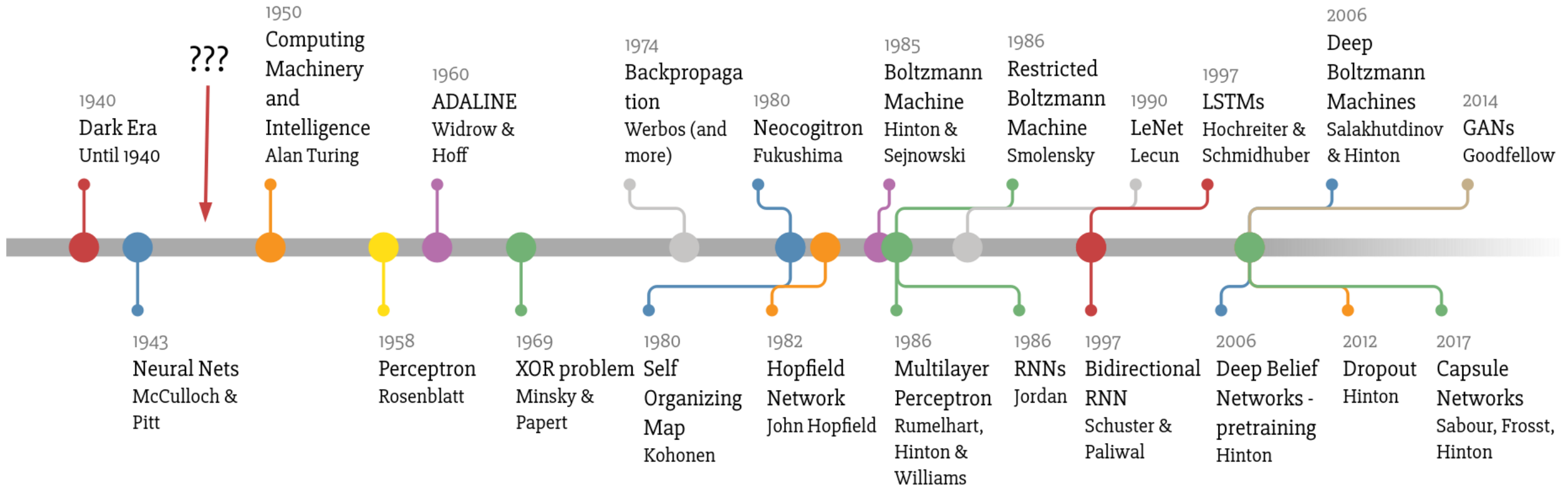


Image from: [link](#)

Course Overview

Rough overview of what we'll cover

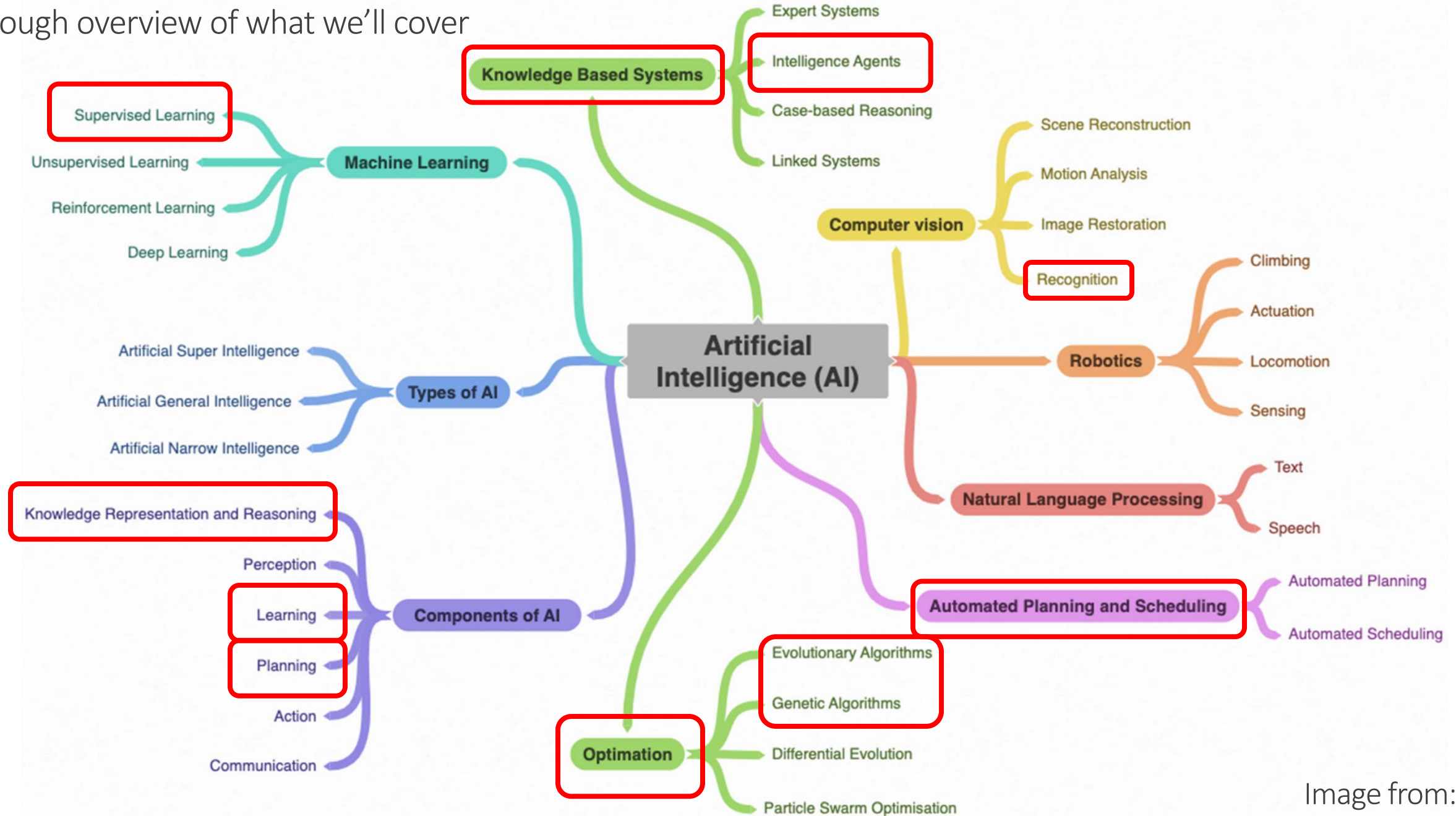


Image from: [link](#)

Rough overview of what we'll cover

- **Searching:** given an environment and some destination, **computationally search** and propose a path.
- **Constraint-satisfaction (optimization):** given an environment and some conditions, **computationally search** and find a state of the environment that satisfies the condition.
- **Game playing:** how to operate in 2-player (or multiplayer) environments.
- **Knowledge-based agents:** develop agents that can answer questions about the environment it is in based on the knowledge (stored in it) it has so far.
- **Machine Learning (supervised learning):** develop programs that can accurately make a decision on a specific task. For example, predict market price of a new house given its properties.
- **Computer Vision (Image Processing):** image processing techniques using filters.
- **Automated Planning:** the main components of general-problem automated planning.
- **Reinforcement Learning:** develop agents that are able to learn how to perform well in an environment through taking actions and receiving feedback. Given enough time the agent would learn the best policy in an environment.

Summary

Summary

- Define intelligence and its characteristics.
- Define artificial intelligence and identify its foundations.
- The standard model of AI is concerned with building agents that **act rationally** to maximize their expected reward.
- Identify domains and applications for AI.
- Learn a bit of history of AI.